TOHID BIJALIKHAN

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Portfolio: Open my VLSI portfolio

SUMMARY

Electronics Engineering graduate specializing in VLSI Design and Technology, with hands-on exposure to RTL design, functional verification, FPGA implementation, CMOS circuit design, and IC layout. Experienced with Verilog/SystemVerilog, Cadence Virtuoso, and ModelSim through internship and academic projects. Seeking entry-level opportunities in VLSI, Digital Design, RTL, FPGA, or semiconductor engineering

EXPERIENCE

VLSI/FPGA Intern

Semitec Electronics India Pvt. Ltd

02/2026 - 05/2026 Jigani Industrial Area, Bengaluru, Karnataka

4-bit Arithmetic Logic Unit (ALU) | Verilog, RTL Design, Testbench, Simulation, FPGA

- Designed and verified a 4-bit ALU at RTL level and progressed the design from simulation to FPGA implementation
- Developed the ALU in Verilog RTL to implement arithmetic and logical operations
- Created a testbench and performed simulation-based functional verification to validate the design across input combinations
- Progressed the verified RTL through the FPGA implementation flow, gaining hands-on exposure to taking a digital design from RTL to hardware

EDUCATION

B.E. in Electronics Engineering (VLSI Design and Technology)

KLE Technological University, Hubballi

2022 - 2026

CGPA 7.66 / 10.00

SKILLS

Programming

Verilog SystemVerilog C Python

VLSI / EDA Tools

Cadence Virtuoso ModelSim

Core Areas

Digital Electronics RTL Design Functional Verification

CMOS/IC Layout FPGA Implementation

CUSTOM

- ICT4SD 2025
- Coursera 2025
- NPTEL

PROJECTS

SRAM Cell Design (6T-10T)

Cadence Virtuoso, CMOS, SRAM, DRC/LVS

- Designed and analyzed 6T-10T SRAM cells to study stability and circuit behavior
- Performed schematic simulation and stability analysis of different SRAM cell configurations
- Designed physical layouts and verified them through DRC/LVS checks in Cadence Virtuoso

NOR1X Standard Cell Design

Cadence Virtuoso, CMOS, Standard Cell, DRC/LVS

- Implemented and verified a NOR1X standard cell with focus on layout correctness and area/performance trade-offs
- Developed the cell through schematic and physical layout implementation
- Performed DRC/LVS verification and worked on layout optimization for improved area efficiency and performance

Standard Cell Layout Design

Cadence Virtuoso, CMOS Layout, DRC/LVS

- Designed physical layouts for commonly used logic standard cells
- Created layouts for Inverter, NAND, NOR, AO21, and AO22 cells
- Performed DRC/LVS verification to check design-rule compliance and connectivity

5T Amplifier Design

Cadence Virtuoso, Analog CMOS, Circuit Simulation

- Designed and simulated a 5-transistor amplifier targeting high voltage gain
- Performed transistor-level circuit design and simulation in Cadence Virtuoso
- Achieved approximately 40 dB voltage gain through circuit-level design and analysis

Kicking Goals with AI

YOLO, OpenCV, Computer Vision

- Developed a football-analysis project using computer-vision techniques for automated video/image analysis
- Applied YOLO-based object detection with OpenCV for football-related analysis